



Stereo-vision-based fruiting-shoot diameter estimation using geometric boundary refinement for robotic selective pollination in kiwifruit orchards

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ABSTRACT

Robotic selective pollination enables targeted regulation of fruit distribution in kiwifruit production, for which fruiting shoot (FS) diameter can provide a structural indicator for subsequent load-related decision-making. While vision-based sensing offers a streamlined solution for in-field geometric measurement, it remains constrained by segmentation instability and boundary noise inherent to slender morphology of fine-scale branches. This study proposes a coarse-to-fine diameter estimation framework based on stereo vision perception, which bridges the gap between semantic perception and geometry quantification. Guided by the biological topology of kiwifruit canopies, FS and fruitful cane (FC) instances were segmented by YOLO11x-seg, while branch junction (BJ) was detected by YOLO11x to enhance structural discrimination. A topology-aware modeling strategy was employed to construct FS centerline through skeletonization, establishing a growth-oriented coordinate system that supports fixed-distance sampling from its base. Adaptation to curvature and non-uniform growth was then achieved by constructing local orthogonal cross-sections using Principal Component Analysis-derived tangents, allowing diameter estimation to remain consistent along curved shoot geometries. After that, a volatility-guided boundary refinement strategy was applied to suppress unstable edge fluctuations and over-segmentation artifacts present in the initial segmentation masks. Subsequently, the physical diameter of FS was quantified by converting the Euclidean pixel distance between the paired boundary points into metric units using the local depth information. Experimental results demonstrate significant accuracy improvements, whereas the refinement process reduced mean absolute error (MAE) from 4.44 mm to 1.09 mm compared with direct mask-based diameter estimation. This decrease in error translates into enhanced practical reliability, with the refined diameter accuracy reaching 98.9% within a 3 mm threshold. Statistical analyses further show that averaging measurements across different perspectives reduce random variability, while measurements derived from unrefined masks retain a pronounced systematic positive bias. By enabling shoot-level diameter estimation, this study provides a perceptual foundation for subsequent load-related decision-making in robotic selective pollination of kiwifruit.

1. Introduction

Pollination is a critical biological process determining flower fertilization, yield formation, and final fruit quality in kiwifruit production. Due to limited pollinator activity and unfavorable environmental conditions, natural pollination alone is often insufficient to meet demand in commercial orchards, resulting in unstable yields (Fornaciari et al.,

2025; Ru et al., 2023; Broussard et al., 2023). Assisted pollination by brushes and hand-held sprayers relied on labor-intensive manual operations made contributions to maintain profitability, but these methods are increasingly constrained by labor shortages, rising costs, and limited operational consistency (He et al., 2025; Wu et al., 2022). In response, mechanized and automated pollination systems have attracted growing attention as a promising solution for improving efficiency and ensuring

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stable pollination under large-scale orchards (Eyles et al., 2022; Tacconi et al., 2016). With advances in sensing, control, and computing capacity, robotic pollination has emerged as a promising direction for achieving efficient, consistent, and scalable pollination, shifting the focus from labor substitution toward autonomous agricultural operations (Gao et al., 2023; Williams et al., 2020; Xie et al., 2025).

Selective pollination in robotic systems represents a fundamental shift in kiwifruit pollination from uniform pollen application toward precision-oriented and decision-driven operations. The physiological success of this approach is rooted in the principle that floral quality is non-uniform, specifically, flowers situated in dominant architectural positions or those exhibiting early phenological development possess a superior sink strength and higher final fruit-mass potential (Asadi et al., 2024; Jakopic et al., 2015). According to these biological imperatives, recent robotic pollination systems have integrated agronomic knowledge with engineering techniques to achieve reliable flower detection (Li et al., 2022b; Pan et al., 2024), flowering stage classification (Li et al., 2022a), and spatial localization (Wei et al., 2026). These advances enable selective targeting of flowers with higher yield potential and improve pollen use efficiency. However, effective selective pollination extends beyond target identification and requires quantitative decision-making, specifically determining how many flowers should be pollinated on an individual flower cluster.

The quantitative regulation of fruit load at fruiting shoot level serves as a prerequisite for realizing full potential of selective pollination. Horticultural evidence has shown that excessive pollination of weak branches can lead to small and uneven fruit size and cause competition for resources, while balanced pollination and fruit distribution can improve fruit size, dry matter content and grade ratio leading to yield increase (Patiyal et al., 2024; Lescouret et al., 1998). Among multiple indicators of shoot vigor, fruiting shoot diameter during the flowering stage has been widely recognized as a robust proxy for vascular transport capacity and carbohydrate supply potential (Kulczewski, 2003; Richardson et al., 2021). Previous studies in kiwifruit and other perennial fruit crops have demonstrated strong correlations between shoot diameter and sustainable fruit number, fruit growth dynamics, and final quality attributes (Ahmed et al., 2025; Deng et al., 2023). Consequently, accurate measurement of fruiting shoot diameter is essential for translating selective pollination from flower-level targeting to load-aware decision-making in robotic systems.

Three-dimensional (3D) measurement technologies have been widely investigated as a means of characterizing plant structure and branch geometry in field environments. Reconstruction approaches based on light detection and ranging (LiDAR) scanning, structure-from-motion (SfM), and multi-view stereo (MVS) have been widely applied to plant modeling by generating high-resolution point clouds that capture complex canopy architectures and support various phenotyping tasks (Wang et al., 2026; Shimbo et al., 2026; Fei et al., 2025; Sumnall et al., 2025). Sun et al. (2024) estimated the morphology of primary branches in apple trees during deciduous period by LiDAR scanning combined with PointNet++ instance segmentation, achieving a branch count accuracy of 93.64% and a mean absolute percentage error of 12.00% in branch length. Mach et al. (2025) developed a 3D reconstruction platform by SfM-MVS-based photogrammetry for plant phenotyping (tomato, lettuce, and cucumber) in controlled laboratory, which reported a root-mean-square reprojection error in pixels of 0.1–0.22 with a scanning time of 2.7 min. Although numerous efforts have achieved, high system cost and substantial computational demands for dense data processing constrain their economic feasibility in practical deployment. Additionally, these approaches often struggle with point-cloud adhesion phenomenon and textures absent in unstructured field conditions (Song et al., 2025; Wang et al., 2025a, 2025b), which may reduce the accuracy of fine measurements of thin shoots under reflective leaves and variable lighting conditions.

Vision-based measurement approaches that combine object segmentation with depth information offer a more lightweight and flexible

alternative for branch and shoot characterization. Monocular red-green-blue (RGB) systems excel at instance segmentation and semantic understanding yet lack inherent scale information, making absolute measurements unreliable without additional priors (Shen et al., 2025; Wu et al., 2019). Depth-enabled vision systems overcome this limitation by acquiring metric depth through active sensing approaches, such as time-of-flight (ToF) and structured light (SL), or through passive stereo vision (SV) (Condotta et al., 2020). Among these approaches, SV is particularly suitable for orchard deployment because of its passive sensing mechanism, relatively low hardware cost, and adaptability to field illumination conditions (Tang et al., 2020). Recent studies have combined stereo depth with instance segmentation to enable geometric measurement in plants, forming a common pipeline of segmentation, depth association, and dimension estimation (Oh and Lee, 2025). Nevertheless, direct boundary-based measurement from segmentation masks remains problematic for thin and deformable structures such as fruiting shoots. Small errors in boundary extraction and depth discontinuities have limited influence on large objects but are significantly amplified for millimeter-scale structures (Ahmed et al., 2025; Yao et al., 2025). These limitations indicate the need for a specialized measurement strategy that explicitly accounts for the structural properties of fruiting shoots and the stability of local geometry.

In this study, an fruiting shoot diameter estimation framework with SV was proposed, which integrates instance segmentation, skeleton-guided measurement localization, and boundary refinement to support kiwifruit selective pollination in robotic systems. Branches of fruitful cane (FC) and fruiting shoot (FS) in kiwifruit canopy were segmented to distinguish instances and their attributes, while branch junctions (BJ) were particularly detected to enhance structural discrimination. The growth direction of each shoot is then reconstructed through skeletonization, and a biologically meaningful measurement point is adaptively determined using depth interpolation to ensure consistent physical spacing from the base. Initial diameter boundaries are obtained from mask-constrained intersections, followed by applying a stability-guided optimization strategy to suppress the influence of segmentation noise, edge ambiguity, and local curvature. By enabling reliable fine-scale geometric perception under complex field conditions, the proposed method provides shoot-diameter information that can serve as a perceptual input for subsequent load-related decision-making in selective pollination.

2. Materials and methods

The proposed framework follows a hierarchical processing strategy for fine-scale FS diameter estimation. The workflow consists of three stages, including perception and reconstruction, topology and localization, and boundary refinement and estimation. Binocular images are first processed to obtain branch perception results and depth information. These outputs are then integrated to establish the oriented FS topology and determine the local measurement cross-section. Finally, the initial mask-based boundaries are used to obtain the preliminary boundary and define the refinement region, from which the refined diameter is derived after local boundary refinement. The overall technical workflow is illustrated in Fig. 1, with the detailed procedures described in the following subsections.

2.1. Data acquisition

Data collection was performed during the flowering stage, a phenological window that corresponds to the critical period for selective pollination and early fruit load regulation. All data were obtained in the International Kiwifruit Innovation Orchard, Yangling, Shaanxi Province, China (34°17' N, 108°2' E; approximately 504 m altitude) in May 2025, where vines were cultivated under a pergola-based trellis system with standardized row spacing and canopy management practices. A consumer binocular camera (JIERUIWEITONG SH500) with a resolution of

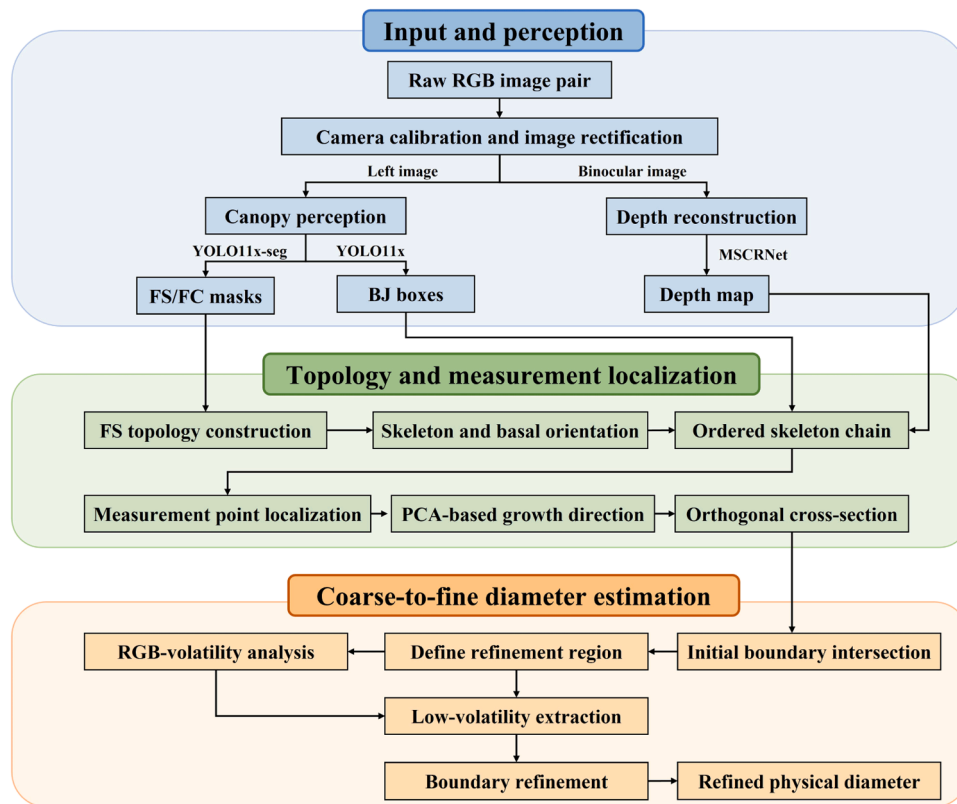


Fig. 1. Workflow of proposed framework for FS diameter estimation.

2560 × 1440 pixels and a 100° field of view (FoV) was utilized to acquire left-right RGB image pairs. The camera was mounted on a mobile platform at a fixed height of approximately 50 cm below the canopy and oriented perpendicular to the canopy plane. All images were acquired under natural field illumination, including sunny, cloudy, and partially shaded conditions, to capture realistic variations in light, occlusion, and background complexity (as shown in Fig. 2). After excluding images with severe motion blur and extreme overexposure that precluded manual annotation, a total of 400 pairs of images were selected for branch segmentation and junction detection. All images are available at https://github.com/Smart-Ag/Kiwifruit_branch_estimation_images.

Ground-truth diameters of fruiting shoots were obtained through in situ manual measurements on living shoots in the orchard to rigorously validate the proposed estimation framework. Measurements were performed at a fixed distance of 10 mm from the shoot base using a Meinaite MNT919310 digital caliper (Meinaite, Germany) with a measurement range of 0 ~ 150 mm, a resolution of 0.01 mm, and a manufacturer-stated accuracy of ± 0.02 mm. The caliper jaws were maintained perpendicular to the local longitudinal axis of the shoot. Considering that the cross-section of a living fruiting shoot may deviate from a perfect circle, three measurements were taken at different randomly selected circumferential orientations around the shoot axis. The arithmetic mean of the three measurements was recorded as the ground-truth diameter (D_T). For isolated shoots with unobstructed

bases, each shoot was associated with a single image, resulting a total of 94 diameter samples. As the FoV of perception system is substantially larger than the effective operating region of a single flower cluster, specific shoots could be observed from different perspectives during target pollination. An additional 25 shoots were measured for the emulation of this multi-perception process, with each set of five shoots associated with 5 images captured from slightly different viewpoints, including the target shoot base and regions beneath adjacent shoots. Each physical measurement was manually aligned with its corresponding image region to ensure accurate spatial correspondence.

2.2. Dataset preparation

High-precision manual annotation forms the foundation for supervising branch segmentation and junction detection tasks. Prior to image annotation, the binocular camera was calibrated using the Stereo Camera Calibrator toolbox of MATLAB R2025a (MathWorks, 2025) to acquire the intrinsic and extrinsic parameters utilized for the rigorous distortion correction and epipolar rectification of collected raw image pairs. Based on these corrected left images, high-precision manual annotation was performed to supervise the branch segmentation and junction detection tasks. All annotations were generated following a unified annotation protocol to ensure consistent interpretation of branch structures under complex canopy conditions. Two complementary annotation strategies were implemented to address the distinct requirements of instance segmentation and object detection. For branch-level structural representation, FS and FC were annotated using polygon-based masks with ISAT (https://github.com/yatengLG/ISAT_with_segment_anything). The annotation criteria were defined according to explicit morphological and topological characteristics of kiwifruit branches. To maintain geometric accuracy, only FS instances featuring a visible base and a distinct structural connection to a parent FC were annotated. The scope of these polygon annotations was traced strictly from the shoot base to the first occlusion



Fig. 2. Examples of kiwifruit canopy images captured in a field orchard under varying illumination conditions of (a) backlighting, (b) front lighting, and (c) overcast conditions.

boundary. For partially occluded FC, only the visible region was annotated without extrapolating hidden structures. This approach enables accurate delineation of irregular branch boundaries under complex canopy backgrounds while preserving fine geometric details for pixel-level supervision. In parallel, branch junction (BJ), which serves as critical cues for growth direction inference and measurement localization, was annotated with axis-aligned bounding boxes by LabelImg (<https://github.com/tzutalin/labelImg>). Each bounding box enclosed a visually discernible junction between a FS and its supporting FC, ensuring consistent spatial correspondence across samples. Representative annotation examples were shown in Fig. 3. The detailed image and instance distributions for the different experimental subsets are summarized in Table 1, with the diameter-evaluation images collected independently of the 400 stereo image pairs used for perception model development.

Considering the complexity of the unstructured orchard environment and the relatively limited number of raw images, data augmentation was applied exclusively to the training set to increase sample diversity and reduce the risk of overfitting. An identical partition index was applied to the raw image-label pairs for both instance segmentation and object detection, randomly dividing the dataset into training and validation sets at a ratio of 8:2. This synchronized partition ensured consistent data separation across the two perception tasks and prevented information leakage between the training and validation subsets. Geometric transformations, including horizontal and vertical flipping and rotations of 90°, 180°, and 270°, were applied to the training images to increase variations in branch orientation. The validation set remained unaugmented to preserve an independent basis for model evaluation.

Table 1

Summary of data used for perception model development and FS diameter evaluation.

Task	Objects	Raw images	Instances
Model training	FC	320	1858
	FS		3162
	BJ		3069
Model validation	FC	80	483
	FS		719
	BJ		872
GT of FS in single-view	FS	94	94
GT of FS in multi-view	FS	25	25

2.3. Canopy perception based on deep learning

2.3.1. Network selection

Both the instance segmentation and object detection tasks were implemented using the You Only Look Once version 11 (YOLO11) series (<https://github.com/ultralytics/ultralytics>), specifically YOLO11x-seg for branch segmentation and YOLO11x for junction detection. YOLO11 adopts an anchor-free paradigm inherited and refined from YOLOv8, and employs a backbone composed of optimized C3k2 blocks, which replace the C2f modules to improve computational efficiency and gradient propagation. A Spatial Pyramid Pooling–Fast (SPPF) module is integrated to expand the effective receptive field, followed by a Cross Stage Partial with Spatial Attention (C2PSA) module that enhances spatial feature selectivity (Khanam and Hussain, 2024). To address the pronounced scale variability of canopy structures, the network adopts an enhanced neck with bidirectional multi-scale feature fusion. Furthermore, YOLO11 retains a decoupled head design that separates classification and regression objectives, improving training stability and

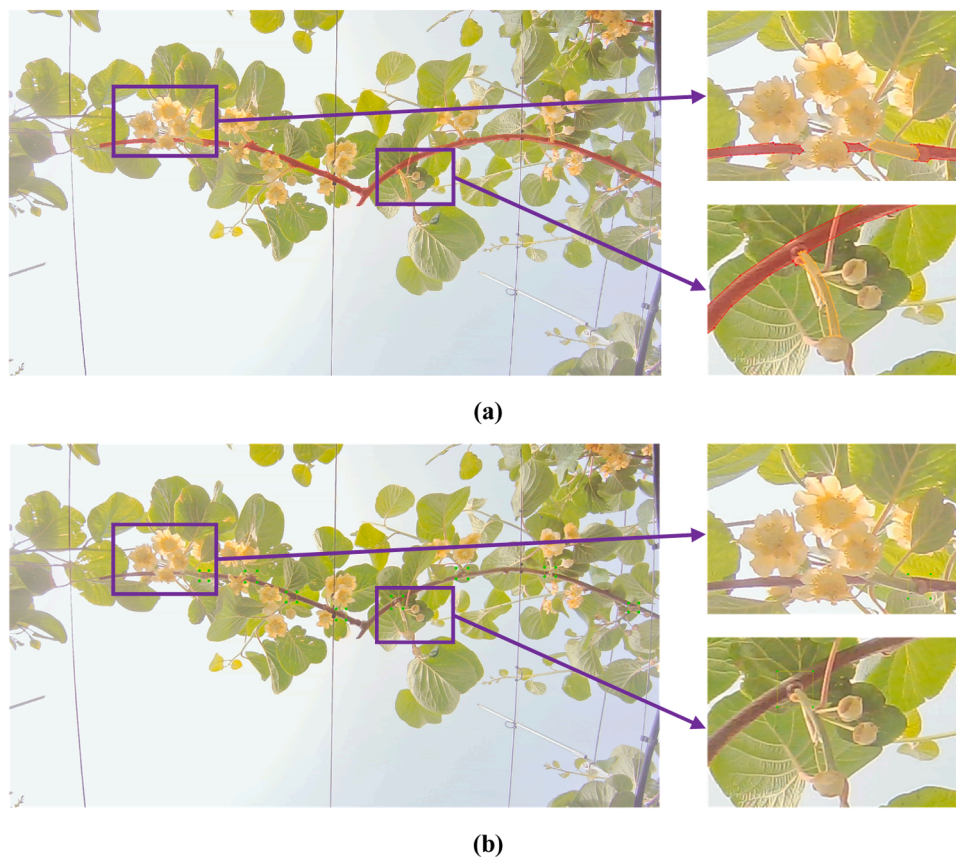


Fig. 3. Examples of multi-class annotation for kiwifruit canopy, including (a) instance segmentation labels of FS (orange polygons) and FC (red polygons), (b) object detection labels of BJ (green rectangles).

localization accuracy. For the segmentation task, an additional parallel branch is introduced to predict prototype masks and corresponding coefficients, allowing precise pixel-level delineation while maintaining real-time inference capability. Compared with lightweight variants, the x-scale configuration provides greater representational capacity, offering potential advantages in preserving structural continuity and local boundary information for slender FS under complex canopy conditions. Since the segmented masks directly provide the geometric basis for subsequent diameter estimation, the present study prioritizes fine-scale perception quality over minimizing network complexity.

2.3.2. Network training

The training and validation of proposed models were executed on a dedicated high-performance workstation to ensure strict experimental standardization. The hardware configuration was anchored by an Intel Xeon Gold 5218 CPU, which was paired with 128 GB of RAM and an NVIDIA RTX 4090 GPU containing 24 GB of VRAM. The software environment operated within a 64-bit Windows 10 system, where the algorithms were deployed using PyTorch 2.2.2 deep learning framework (Python 3.10.14) supported by CUDA 12.1 and cuDNN 8.8.1 acceleration. Considering the relatively limited scale of the orchard dataset, transfer learning was adopted rather than training the networks from random initialization. The model parameters were initialized using weights (<https://docs.ultralytics.com/zh/models/yolo11>) pretrained on the large-scale Common Objects in Context (COCO) dataset and subsequently fine-tuned on the kiwifruit orchard training set. In addition to the geometric transformations applied during dataset preparation, the online augmentation mechanisms integrated into the YOLO training framework were enabled during model optimization. These included Mosaic augmentation, photometric variation, random translation and scaling, and probabilistic horizontal flipping, thereby increasing the diversity of object appearances, scales, spatial arrangements, and canopy backgrounds encountered during training. Consistent training parameters included an input size of 1280, a batch size of 4, and a commencing learning rate of 0.0001. The maximum training epochs were set to 500 as an upper limit, while an early-stopping mechanism with a patience of 50 epochs was employed to terminate training automatically when the validation performance ceased to improve.

2.4. Depth estimation

Depth information was computed from disparity maps generated via stereo matching on the rectified binocular image pairs. For robust dense matching in unstructured orchard environments, the multi-scale structure-guided and cascaded disparity refinement network (MSCRNet) proposed by Jing (2025) was utilized. Technically, MSCRNet employs a coarse-to-fine architecture anchored by a MobileNetV2 backbone. It integrates a novel structure-convolutional module that explicitly encodes pixel-wise geometric relationships, which effectively mitigate texture ambiguity common in canopy foliage. Initial disparity is regressed from a group-wise correlation cost volume and iteratively optimized through a cascade refinement mechanism. This process utilizes warping-based cost volumes and Squeeze-and-Excitation attention layers to suppress background noise and enhance boundary precision. To ensure strong domain adaptability, the model used for stereo matching was trained on the domain-specific KiwiStereoDataset (KS-Dataset) that comprises of field-acquired stereo pairs and

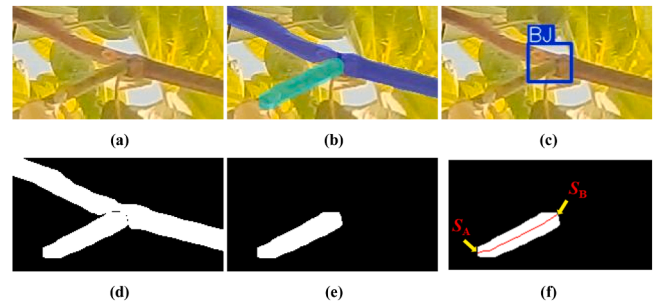


Fig. 4. Process of skeleton extraction and oriented topology establishment. (a) Branch instance; (b) Segmentation results of FS (indigo polygons) and FC (blue polygons); (c) Detection results of BJ (blue rectangle); (d) Binary mask of segmented branch; (e) Binary mask of segmented FS instance; (f) Skeleton extraction of segmented FS instance with the identified basal node S_B and apical tip S_A .

ground-truth disparity maps generated via an iterative two-stage LiDAR filtering algorithm (Jing et al., 2024). The pretrained model was directly employed in the present study to generate disparity maps for subsequent depth reconstruction.

After acquiring the high-precision disparity maps by MSCRNet, depth value for each pixel was retrieved based on the standard principles of binocular stereo triangulation. Since input image pairs underwent rigorous stereo rectification, the epipolar lines are strictly horizontally aligned. Consequently, the depth value is inversely proportional to the disparity value, which mathematical relationship is defined in Eq. (1). By applying this transformation pixel-wise, the disparity map was converted into a depth map aligned with the coordinate system of left image. To ensure numerical stability and physical validity, pixels with near-zero disparity values at infinite depths or invalid matches are masked out. The resulting depth map provides essential spatial information required for the subsequent branch diameter estimation.

$$Z = \frac{f \times B}{d} \quad (1)$$

where, Z refers to the depth value from camera plane to target point; f refers to the focal length of left lens to binocular camera; B refers to the baseline of binocular camera; and d refers to the estimated disparity that predicted by MSCRNet.

2.5. Diameter estimation of fruiting shoot

For load-aware selective pollination, a robust and geometry-consistent diameter estimation algorithm was developed for fruiting shoots under complex canopy conditions. Unlike conventional width measurement methods that rely on direct mask projection or single-line fitting, the proposed approach integrates instance-level segmentation, skeleton-guided localization, depth-aware metric conversion, and boundary refinement to ensure both spatial consistency and physical interpretability. Specifically, the computational procedure initiates with skeleton extraction and growth direction determination to establish a local coordinate system for measurement. Subsequently, the workflow advances to orthogonal scanline generation and boundary localization to identify candidate edge points in the image plane. Finally, the paired boundary points are determined through local boundary refinement,

and their image-plane distance is converted into the physical diameter using the corresponding local depth information. The pseudocode of proposed framework is formally outlined below.

2.5.2. Skeleton-guided localization of measurement cross-section

Adhering to horticultural measurement standards that require diameter quantification at a fixed physical distance from S_B , this step necessitates an accurate conversion from physical distance to cumula-

Input:
 RGB-D image pair of I_{RGB} and I_{Depth}
 Instance masks of FS $\{M_{FS}\}$ and FC $\{M_{FC}\}$ segmented in I_{RGB} by YOLO11x-seg
 Boxes of BJ $\{B_{BJ}\}$ detected in I_{RGB} by YOLO11x
 Base line and focal length of left lens from parameter of binocular camera

Output:
 Primary diameter $\{D_P\}$ and refined diameter $\{D_R\}$ of segmented $\{M_{FS}\}$

S1: **For** each M_{FS} and M_{FC} **Do**
 S2: Preprocess all masks $\{M_{FS}\}$ and $\{M_{FC}\}$ by retaining the largest connected component
 S3: Merge overlapping masks of $\{M_{FS}\}$ and $\{M_{FC}\}$
 S4: **End for**
 S5: **For** each M_{FS} **Do**
 S6: Extract the topological centerline S through morphological thinning
 S7: Identify the proximal basal node S_B and apical growth tip S_A of S based on spatial proximity
 S8: Construct an ordered skeleton chain starting from the determined base point S_B
 S9: Estimate a preliminary measurement point S_j at a target distance using global median depth Z_G of M_{FS}
 S10: Estimate the refine measurement point S_M by a second depth-aware interpolation
 S11: Extract the neighborhood skeleton of S_M
 S12: Estimate the local growth direction by PCA and compute the normal direction v
 S13: Perform a bilateral search along v from S_M , find initial mask intersections of E_A and E_B
 S14: Calculate the Euclidean length between E_A and E_B as D_P
 S15: Extend the boundary segment geometrically by a ratio of α ($\alpha = 0.2$)
 S16: Sample multi-channel values along the extended segment
 S17: Identify the most stable middle segment using low-volatility criteria
 S18: Set the endpoints of stable segment as refined boundary intersections of $E_{A'}$ and $E_{B'}$
 S19: Calculate the Euclidean length between $E_{A'}$ and $E_{B'}$ as D_R
 S20: **End for**
 S21: **Return** $\{D_P\}$ and $\{D_R\}$

2.5.1. Skeleton extraction and oriented topology establishment

A topology-aware skeleton extraction strategy is adopted to convert binary instance masks into ordered centerlines with biologically meaningful orientations. The procedure begins with a morphological filtering stage, in which the largest connected component of each segmented FS and FC is retained to remove isolated fragments and disconnected noise artifacts before skeletonization (as shown in Fig. 4(a), Fig. 4(b), and Fig. 4(d)). Subsequently, a skeletonization algorithm based on the medial axis transform is applied to the processed FS masks (M_{FS}), yielding a one-pixel-wide centerline, denoted as S (as shown in Fig. 4(e)). Within this topological representation, the endpoints are identified as skeleton extremities, comprising the basal node (S_B) connected to FC and the apical growth tip (S_A). A hierarchical multi-source fusion strategy integrating object detection priors with semantic segmentation cues is proposed to determine growth-oriented structure of FS. Specifically, if an endpoint spatially overlaps with a detected bounding box of BJ (B_{BJ}), it is assigned as S_B (as shown in and Fig. 4(f)). If no such overlap is found, the supporting FC is used as an additional structural cue. The spatial proximity between each skeleton endpoint and the FC is then evaluated, and the endpoint with the minimum Euclidean distance to the FC is assigned as S_B . Once the biological attributes of endpoints in S were determined, the unordered skeleton pixels are reorganized into a directed graph originating from S_B (as shown in Fig. 4(f)), forming a strictly ordered skeleton chain, denoted as C_K .

tive pixel path length. Considering depth fluctuations caused by stereo matching errors in unstructured orchard, relying on a static scaling factor may introduces substantial localization deviations. Consequently, a dual-iterative approximation strategy was developed to dynamically locate the optimal measurement site along C_K . The process begins with a coarse estimation to establish a region of interest that approximates the location of the target measurement point. Based on the pinhole camera model, the physical length (L_{phy}) is converted into pixel path length (L_{px1}) by Eq. (2), where the global median depth (Z_G) is extracted from C_K to represent Z and mitigate the influence of outliers. Subsequently, traversing C_K from S_B , the cumulative skeletal distance $SD(j)$ at index j is obtained by Eq. (3), defined as the sum of Euclidean distances between consecutive skeleton nodes. A preliminary measurement point S_j is identified as the first point satisfying $SD(j) \geq L_{px1}$.

$$L_{px} = \frac{L_{phy} \cdot f}{Z} \quad (2)$$

$$SD(j) = \sum_{i=0}^{j-1} \|S_{i+1} - S_i\|_2 \quad (3)$$

where, L_{px} refers to the pixel path length; L_{phy} refers to the physical length of shoot from its base, which was set to 10 mm that maintain consistency with D_T acquisition; S_i represents the pixel coordinates of the i th point form C_K ; $\|\cdot\|_2$ refers to the Euclidean norm in 2D pixel coordinate system.

As the depth of slender shoots exhibits gradual variation along their growth direction, the fidelity of physical-to-pixel mapping based on a global depth prior inevitably degrades with increasing traversal dis-

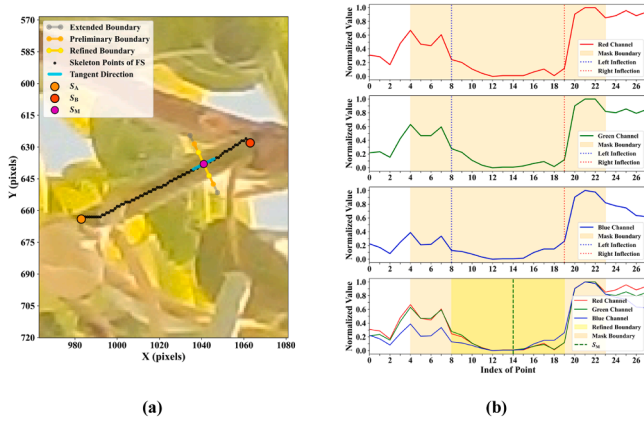


Fig. 5. Schematic illustration of the coarse-to-fine diameter estimation. (a) Skeleton-guided measurement point localization and establishment of the growth-oriented coordinate system for orthogonal sampling; (b) Boundary refinement by multi-channel intensity extraction.

tance. To alleviate this cumulative bias, the median depth of all skeleton nodes between S_B and S_j is recalculated and denoted as the local reference depth Z_L . Consequently, the target pixel path length is updated to L_{px2} by Eq. (4). With refined path length, the skeleton chain is retraversed to determine the final measurement point of S_M by Eq. (4). Subsequently, a local skeletal neighborhood (5-pixel window) centered at S_M is extracted to capture local geometric trends while reducing sensitivity to contour noise and skeleton perturbations. Principal Component Analysis (PCA) is then applied, where the dominant eigenvector of the covariance matrix is taken as the local tangent vector. The corresponding normal direction v is obtained through orthogonal rotation of the tangent vector, defining the measurement cross-section for diameter estimation. The resulting normal direction serves as the reference axis for subsequent boundary localization.

$$L_{px2} = L_{px1} \cdot \frac{Z_G}{Z_L} \quad (4)$$

2.5.3. Orthogonal boundary refinement and diameter estimation

Once the measurement cross-section has been determined, the shoot diameter can be estimated by identifying boundary intersections along the normal direction. A discrete scanline is constructed by sampling bidirectionally from S_M along v . By traversing the binary segmentation mask of M_{FS} , the first foreground-to-background transitions on both sides are identified, achieving a pair of preliminary boundary intersections, denoted as E_A and E_B . These two points define the preliminary spatial extent of the shoot at a given cross-section, providing the basis for initial diameter estimation. To account for local depth variation during metric conversion, the median depth of the pixels between E_A and E_B along the scanline is adopted as the representative depth Z_L , reducing sensitivity to isolated depth fluctuations by MSCRNNet. By aggregating depth information across the local cross-section rather than relying on a single-pixel estimate, the resulting scale factor is less susceptible to local disparity instability. The FS diameter is quantified by projecting Euclidean distance between E_A and E_B from image plane into physical space using Eq. (5), which is recorded as the preliminary estimated diameter D_p . While this procedure offers a direct and computationally efficient approximation, the resulting boundary intersections remain susceptible to discretization artifacts, local segmentation noise, and contour irregularities.

$$D_E = \|E_B - E_A\|_2 \cdot \frac{Z_L}{f} \quad (5)$$

where, D_E refers to the estimated diameter of FS; Z_L refers to the median depth of all points between E_A and E_B alone constructed

scanline.

Addressing these inherent precision limitations, a boundary refinement strategy integrating geometric extension and multi-channel stability analysis is implemented to suppress edge uncertainty and segmentation artifacts. The orthogonal scanline is first symmetrically extended by a proportional factor of α ($\alpha = 0.2$), forming an expanded candidate interval designed to encompass potential boundary displacement caused by both over-segmentation and under-segmentation while limiting the inclusion of irrelevant background regions (as shown in Fig. 5(a)). Along this extended line, the red, green, and blue intensity values of each pixel are extracted to construct a multi-channel feature representation of the surface texture characteristics. For each channel, a sliding window is applied to compute the local feature volatility by Eq. (6), which transforms the binary, single-channel boundary problem into a multi-channel stability analysis task. As illustrated in Fig. 5(b), homogeneous interior regions of the shoot correspond to low-volatility intervals, whereas shoot-background transitions and residual background pixels introduced by segmentation uncertainty induce sharp volatility responses. Endpoints of the longest contiguous low-volatility interval are selected as candidate boundary intersections according to Eq. (7). Since the stable intervals obtained from different channels may not completely coincide, the final aggregated stable region with refined boundary intersections of E_A' and E_B' is through an envelope operation across all channels. Based on these optimized intersections, the refined diameter D_R is computed by projection the Euclidean distance between E_A' and E_B' into the physical metric space by Eq. (5).

$$\sigma_c(i) = \frac{1}{\omega} \sum_{t=0}^{\omega-1} |I_c(i+t) - I_c(i+t-1)| \quad (6)$$

$$R_c = \underset{[t_1, t_2]}{\operatorname{argmax}} \{ \{ (t_2 - t_1) | \forall t \in [t_1, t_2], \sigma_c(t) < \beta \} \} \quad (7)$$

where, $\sigma_c(i)$ refers to volatility score for channel c at point index i ; ω refers to the size of adopt sliding window, which is set to half the number of pixels from E_A to E_B ; $I_c(i)$ refers to the intensity value at point index i for channel c ; R_c refers to the optimal stable interval for channel c ; t refers to the arbitrary point index traversing within the candidate interval $[t_1, t_2]$ for channel c ; β refers to the adaptive stability threshold for channel c , which was set to 0.3 of the volatility distribution and applied consistently to all samples throughout the evaluation.

2.6. Evaluation metrics

Two categories of metrics were adopted to quantitatively evaluate the proposed framework, corresponding to the deep learning-based models and shoot diameter estimation. Performance of instance segmentation and BJ detection was assessed using Precision (P), Recall (R), Average Precision at an Intersection over Union (IoU) threshold of 0.50 (AP_{50}), and the corresponding mean Average Precision across classes (mAP_{50}). AP_{50} is calculated as the area under the Precision-Recall (PR) curve at IoU = 0.50, with the PR curve generated by varying the confidence threshold. The reported P and R correspond to the operating point associated with the maximum mean F1 score on the validation set. For YOLO11x-seg, these metrics were calculated at the mask level, whereas bounding-box-level metrics were used for BJ detection. These metrics jointly characterize both localization accuracy and classification reliability provided in Eq. (8) to Eq. (11).

$$P_n = \frac{TP_n}{TP_n + FP_n} \quad (8)$$

$$R_n = \frac{TP_n}{TP_n + FN_n} \quad (9)$$

$$AP_{50/n} = \int_0^1 P_n(R_n) dR_n \quad (10)$$

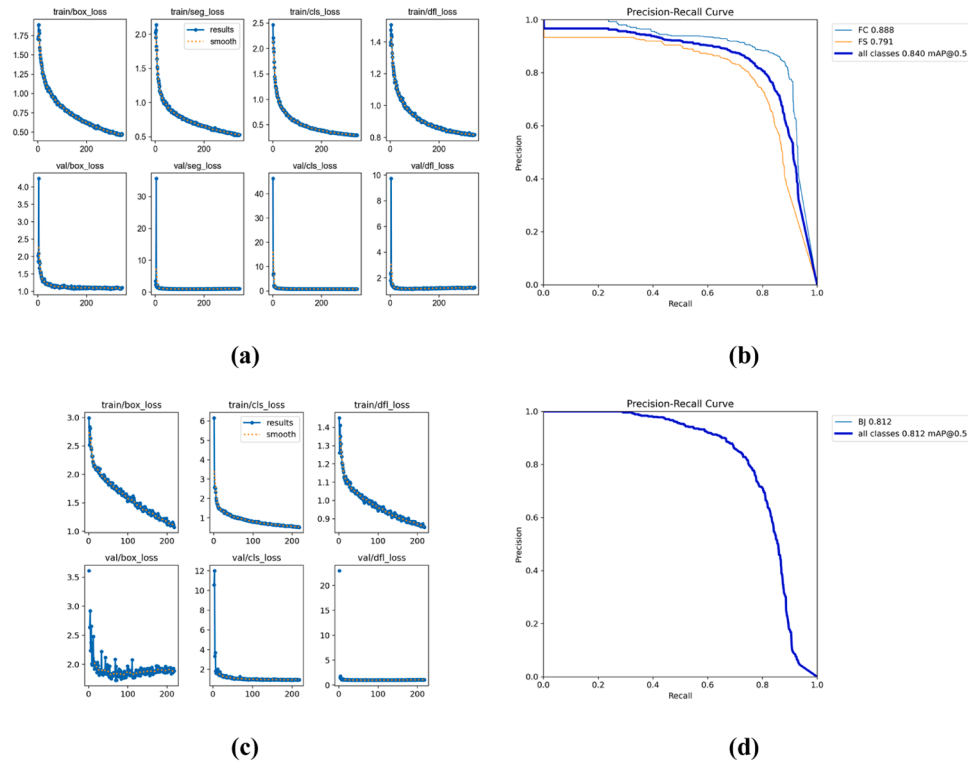


Fig. 6. Training convergence and PR curves of trained (a-b) YOLO11x-seg for branch segmentation and (c-d) YOLO11x for branch junction detection.

Table 2

Performance of trained YOLO11x-seg and YOLO11.

Models	Classes	P	R	AP ₅₀	mAP ₅₀
YOLO11x-seg	All	0.818	0.814	0.840	0.840
	FS	0.793	0.751	0.791	
	FC	0.843	0.877	0.888	
YOLO11x	BJ	0.828	0.736	0.812	0.812

$$mAP_{50} = \frac{1}{l} \sum_{n=1}^l AP_n \quad (11)$$

where, n refers to the index of specific instance class; TP (True Positives) refers to the number of instances within that category accurately identified by model; FP (False Positives) refers to the number of negative instances erroneously assigned as belonging to that class; FN (False Negatives) refers to instance from the class that model failed to identify; and l refers to the total number of instance classes.

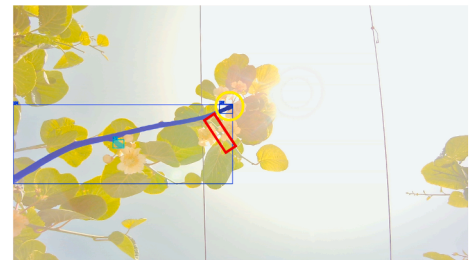
Four evaluation metrics, which included mean error (ME), mean absolute error (MAE), mean absolute percentage error ($MAPE$), and root mean square error ($RMSE$), were employed to quantify the accuracy of estimated FS diameter. ME reflects the systematic bias of estimation, indicating whether the method tends to overestimate or underestimate diameter of fruiting shoots. MAE provides an intuitive measure of the average absolute error, while $MAPE$ enables scale-independent comparison by normalizing errors with respect to the ground truth. $RMSE$ penalizes larger deviations more heavily and is therefore sensitive to outliers. The formulas of these evaluation metrics are provided in Eq. (12) to Eq. (15).

$$ME = \frac{1}{l} \sum_{s=1}^l (D_E(s) - D_T(s)) \quad (12)$$

$$MAE = \frac{1}{l} \sum_{s=1}^l |D_E(s) - D_T(s)| \quad (13)$$



(a)



(b)

Fig. 7. Instance segmentation examples of branches by YOLO11x-seg, where FS and FC were segmented into indigo and blue polygons, respectively. (a) Over-segmentation of FS with noise engaged mask edge; (b) Missed segmentation of FC marked with red rectangle and incomplete segmentation of FS marked with yellow circle under backlighting condition.

$$MAPE = \frac{1}{l} \sum_{s=1}^l \left| \frac{D_E(s) - D_T(s)}{D_T(s)} \right| \times 100\% \quad (14)$$

$$RMSE = \sqrt{\frac{1}{l} \sum_{s=1}^l (D_E(s) - D_T(s))^2} \quad (15)$$

Table 3

Performance of instance segmentation models applied to plant organs in case studies.

References	Plants	Models	Objects	AP	mAP
Ahmed et al. (2025)	Apple tree	YOLOv8x-seg	Trunk Branch	0.969 0.746	0.857
Ji et al. (2025)	Tomato plant	Improved YOLOv8n-seg	Stem	0.793	0.793
Qi et al. (2024)	Tomato plant	Improved YOLOv8s-seg	Bud Stem	0.475 0.855	0.665
Niknejad et al. (2023)	Pine tree	Mask R-CNN	Stem Branch	0.283 0.395	0.339
Sun et al. (2022)	Apple tree	SOLOv2	Trunk Grafting area	- -	0.811
This study	Kiwifruit plant	YOLO11x-seg	Fruiting shoot Fruitful cane	0.791 0.888	0.840

where, s refers to the index of samples; l refers to the total number of samples; D_T refers to the ground truth diameter of FS; and D_E refers to the estimated diameter of FS.

3. Results and discussion

3.1. Performance of branch segmentation and junction detection

The trained deep learning models demonstrated robust perception capabilities in field kiwifruit orchard, effectively distinguishing fine-scale branches and critical growth sites from the complex canopy. The training and validation losses of YOLO11x-seg and YOLO11x decreased rapidly and then stabilized (as shown in Fig. 6(a) and Fig. 6(c)), with early stopping occurring at epoch 346 and epoch 216, respectively, indicating stable convergence of both models. PR curves of two models demonstrate strong detection capabilities with a mAP_{50} of 0.840 for branch segmentation and 0.812 for junction detection, as shown in Fig. 6(b) and Fig. 6(d). Among all classes, FC achieved a higher segmentation accuracy with an AP_{50} of 0.888 (as shown in Table 2), reflecting that the distinct woody texture and larger caliber of the canes facilitated consistent feature extraction. In contrast, FS recorded a slightly lower AP_{50} of 0.791. This performance gap is primarily attributed to the high chromatic and morphological similarity between the emerging fruiting shoots and the background foliage. Specifically, their slender geometry with green color closely resembles that of petioles and pedicels, introducing visual ambiguity that hinders precise boundary definition (Liu et al., 2025). Additionally, for BJ detection, the trained YOLO11x achieved a AP_{50} of 0.812, providing reliable junction cues for subsequent shoot topology construction.

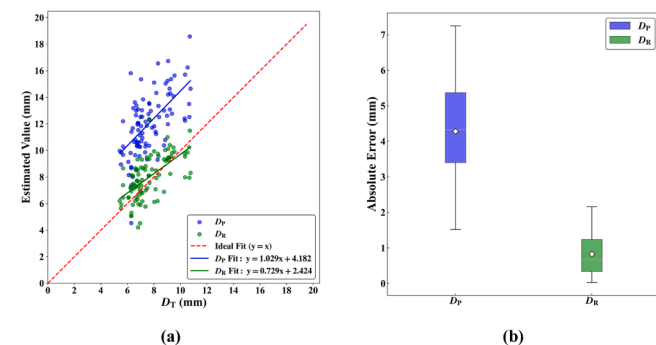


Fig. 8. Result of FS diameter estimation across D_P and D_R , including (a) scatter plot with linear fits (b) box plot of absolute error distribution.

Although the proposed models achieve effective perception to kiwifruit canopy, their accuracy is jointly constrained by environmental complexity and intrinsic morphological properties. Characterized by a slender morphology that occupies a limited number of pixel width and exhibits weak boundary contrast, FS renders mask continuity highly sensitive to illumination variation, consequently resulting in frequent over-segmentation errors (Ji et al., 2025). This issue is further exacerbated in curved shoots, where edge noise emerges along the segmentation contour, as shown in Fig. 7(a). Missed segmentations of FS are also observed under backlighting conditions, where color attenuation and geometric distortion suppress discriminative visual cues. For FC, its elongated and continuous structure increases the risk of incomplete instance extraction, where a single instance may be erroneously divided into multiple segments. As shown in Fig. 7(b), these fragmented masks often spatially overlap and can be consolidated through union-based merging to recover complete cane topology. Despite variations in crop species, datasets, imaging conditions, and evaluation schemes, the reported studies summarized in Table 3 demonstrate that segmentation of slender plant organs remains a challenging task in agricultural environments. Increasing the quantity and diversity of training data has clear potential to further improve segmentation accuracy and generalization. Nevertheless, precise boundary delineation remains challenging for fine-scale plant organs such as stems and branches, which are characterized by slender and curved morphology, limited pixel occupancy, weak boundary contrast, frequent occlusion, and strong visual similarity to surrounding canopy structures. Nevertheless, precise boundary delineation remains challenging for fine-scale plant organs because of their slender and curved morphology, limited pixel occupancy, weak boundary contrast, frequent occlusion, and visual similarity to surrounding canopy structures.

Regarding BJ detection, limitation of trained YOLO11x primarily manifest as redundant overlapping boxes and missed instances. While backlighting contributes to detection failures, a more intrinsic limitation arises from the single-class detection setting that lacks semantic and spatial context with the lowest R of 0.736 among all classes. Integrating multi-class detection, specifically, associating prominent reproductive organs of flowers and clusters with BJ can effectively mitigate this limitation (Li et al., 2022a). After that, the redundancy detections were refined through Intersection over Union analysis. When BJ were entirely missed, the proposed distance-based evaluation strategy was employed as a geometric alternative for the topological construction of FS to infer basal points from the intersection between the skeleton of FS and FC region, ensuring reliable diameter estimation.

3.2. Single-view diameter estimation of FS

The proposed boundary refinement strategy effectively improves the accuracy of FS diameter estimation by correcting segmentation-induced geometric bias. Based on the analysis of 94 samples, D_P exhibits a pronounced positive bias, with a ME of 4.40 mm and an $RMSE$ of 4.77 mm, indicating consistent overestimation rather than random noise. This behavior is clearly illustrated in the scatter plot of Fig. 8(a), where most D_P lie well above the identity line, forming an approximately parallel offset. Consistent evidence is provided by the additional metrics reported in Table 4, including a MAE of 4.44 mm and a $MAPE$ of 60.00% to D_P , further confirming the inflation effect induced by mask boundary distortion. Following boundary refinement, D_R shows a substantial reduction in both bias and dispersion. The ME decreases to 0.36 mm, while $MAPE$ and $RMSE$ are reduced to 14.68% and 1.36 mm,

Table 4

Evaluation of diameter estimation for FS under single-view observation.

Method	Samples	ME (mm)	MAE (mm)	MAPE (%)	RMSE (mm)
D_P	94	4.40	4.44	60.00	4.77
D_R		0.36	1.09	14.68	1.36

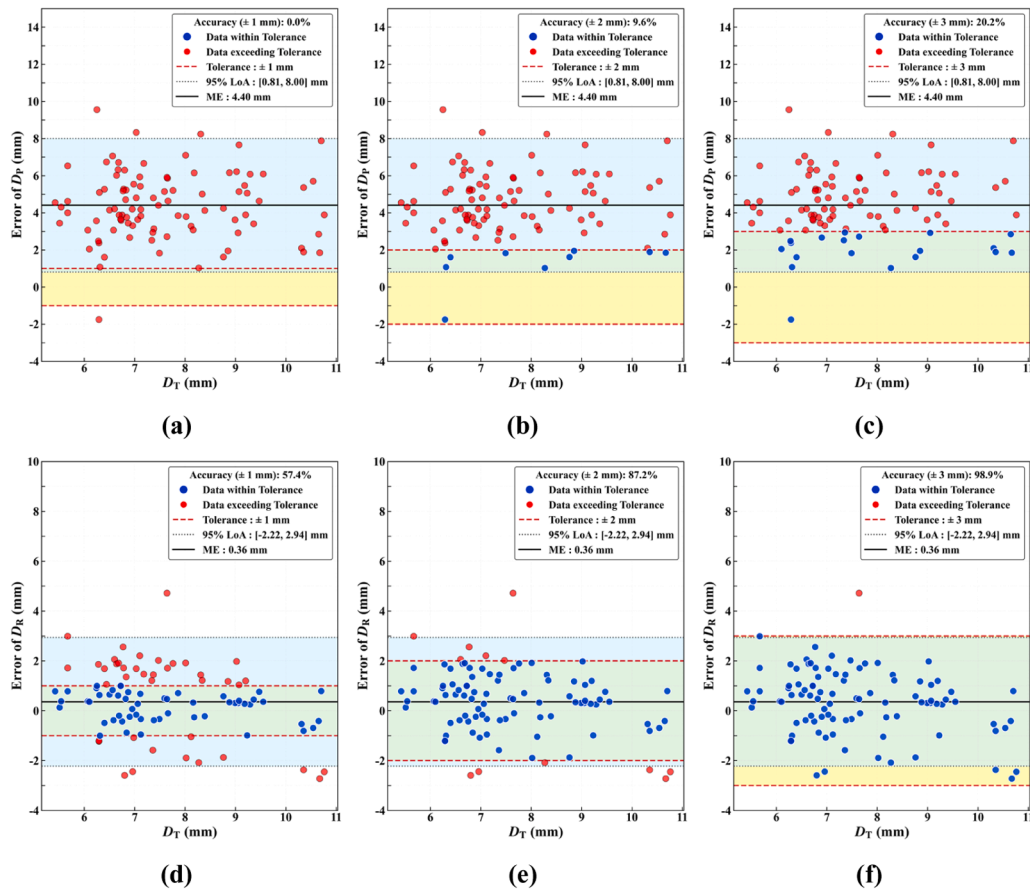


Fig. 9. Tolerance-based accuracy of diameter estimation, where (a)–(c) correspond to D_p under tolerances of 1 mm, 2 mm, and 3 mm, and (d)–(f) correspond to D_R under the same tolerances.

Table 5
Processing time of the proposed diameter estimation framework.

Processing stage		Time
Average processing time per RGB-depth pair (ms/pair)	Junction detection (ms/pair)	55.47
		± 9.89
	Branch segmentation (ms/pair)	76.84
		± 5.62
	Mask post-processing (ms/pair)	140.53
		± 60.99
	Skeletonization and measurement point localization (ms/pair)	336.03
Boundary refinement and diameter estimation (ms/pair)		± 148.62
		14.34
		± 6.41
Total (ms/pair)		623.21
		± 231.47
Average processing time per shoot (ms/shoot)		83.65
		± 31.07

respectively. In the corresponding scatter plot, D_R samples cluster closely around the identity line, demonstrating effective suppression of systematic deviation. In terms of error distribution, D_p exhibits a strongly positively shifted and wider spread, whereas D_R shows a compact distribution centered near zero. As further illustrated in the absolute error box plot of Fig. 8(b), the distribution of D_R contrasts sharply with the wide, positively skewed distribution of D_p . Specifically, the interquartile range of D_R is narrower and centered near zero, with the MAE decreased from 4.44 mm to 1.09 mm.

A supplementary tolerance-based evaluation was conducted to provide an intuitive interpretation of estimation accuracy under varying error thresholds. In this context, the Limits of Agreement (LoA) specify the interval within 95% of the estimation errors fall (95% LoA), whereas

accuracy is defined as the proportion of estimates whose absolute error lies within a prescribed tolerance, illustrated in Fig. 9. From a distribution-level perspective, D_p exhibits a pronounced systematic bias, with a broad 95% LoA spanning from 0.81 mm to 8.00 mm. The wide and positively shifted agreement interval indicates that most errors are dominated by overestimation caused by mask expansion effects, rather than random fluctuations. Consequently, as shown in Fig. 9(a), none of D_p samples fall within a tolerance of 1 mm. This confirms that directly using segmentation masks may be insufficient for fine-scale geometric measurements. In contrast, the D_R recenters the error distribution close to zero, with a substantially narrower 95% LoA from -2.22 mm to 2.94 mm. This indicates that 95% of D_R deviate from D_T by less than 3 mm, demonstrating that the proposed boundary refinement effectively suppresses segmentation-induced systematic noise. Specifically, tolerance-based analysis shows that D_R achieves accuracy of 57.4% within tolerance of 1 mm, 87.2% within tolerance of 2 mm, and 98.9% within tolerance of 3 mm, as shown in Fig. 9(d) to Fig. 9(f). Compared with previous apple branch diameter estimation studies, Chattopadhyay et al. (2016) and Ahmed et al. (2025) reported accuracies of 89.40% and 97.67%, respectively, under a tolerance of 5 mm. In contrast, the proposed method achieves higher accuracy under a stricter 3 mm tolerance, with an MAE of 1.09 mm, which is comparable to the 1.10 mm reported by Tong et al. (2026). Notably, these studies were conducted under

Table 6
Evaluation of diameter estimation for FS under multi-view observation.

Method	Samples	ME (mm)	MAE (mm)	MAPE (%)	RMSE (mm)
D_p	25	4.28	4.28	57.70	4.50
D_R		0.26	0.83	11.16	1.01

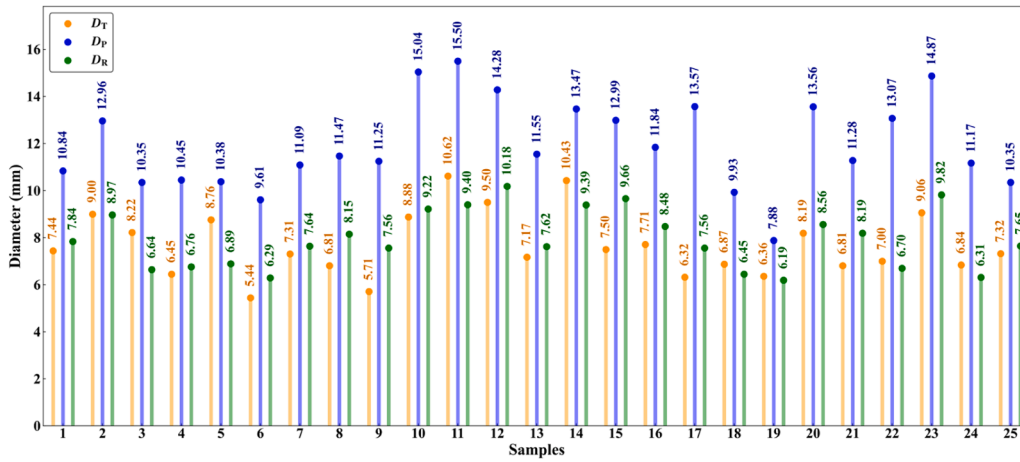


Fig. 10. Lollipop chart of diameter samples of FS with D_T , D_P , and D_R .

dormant-season with low-occlusion conditions, the proposed framework is evaluated during the flowering stage, where dense foliage interference greatly complicates precise geometric measurement. This comparison indicates that the proposed coarse-to-fine visual refinement strategy provides more precise geometric constraints than prior approaches, while remaining effective in field orchard.

The computational efficiency and deployment feasibility of the proposed framework were further evaluated on an edge-computing platform by 40 registered RGB-depth image pairs containing 298 segmented FS (Jetson AGX Orin, NVIDIA, USA). The average processing time for a single image was 623.21 ± 231.47 ms, corresponding to approximately 1.60 FPS (as shown in Table 5). Among all modules, skeletonization and measurement point localization accounted for the largest computational cost of 336.03 ± 148.62 ms, followed by mask post-processing of 140.53 ± 60.99 ms, whereas boundary refinement and diameter estimation required only 14.34 ± 6.41 ms. For an individual FS, the average processing time was 83.65 ± 31.07 ms per shoot. Although the framework adopts the computationally larger x-scale perception models, the measured latency remained compatible with the operational requirements of robotic selective pollination, which does not impose a dominant computational burden on the overall robotic workflow. Compared with the robotic pruning system for apple tree that reported by Tong et al. (2026), which required an average of 6.1 s for pruning-point identification and 17.8 s for complete branch operation on a notebook equipped with an Nvidia GTX3050 GPU, the processing time of the proposed framework represents a relatively small proportion of the overall robotic operation cycle. These results demonstrate the feasibility of deploying the proposed method on edge-computing platforms for field applications.

3.3. Multi-view diameter estimation of fruiting shoots

Integrating observations from multiple-view reduces errors caused

Table 7

Statistical analyses of multi-view diameter-estimation errors.

Error analysis items	Methods	
	D_P	D_R
Shapiro-Wilk test	$W = 0.98$ $p = 0.91 (> 0.05)$	$W = 0.97$ $p = 0.73 (> 0.05)$
Wilcoxon signed-rank test	$W = 0$ $p = 5.96 \times 10^{-8} (***)$	$W = 109$ $p = 0.16 \times 10^{-3} (ns)$
One-sample t -test	$p = 9.09 \times 10^{-14} (***)$ $t(24) = 15.12$	$p = 0.21 (ns)$ $t(24) = 1.28$
Paired t -test of absolute errors	$t(24) = 11.52, p = 2.90 \times 10^{-11} (***)$	
Mean absolute-error reduction (mm)	3.45 (95% CI: [2.83, 4.07])	

Note: ***, $p < 0.001$; ns, not significant.

by projection ambiguity and local occlusions, enhancing the reliability of FS estimation. For recorded 25 diameter samples of FS, the final estimated diameter was derived by calculating the arithmetic mean of all available views. This multi-view averaging systematically decreased errors for both D_P and D_R , as summarized in Table 6. Specifically, multi-view averaging reduced the RMSE of D_P from 4.77 mm to 4.50 mm, and further improved that of D_R from 1.36 mm to 1.01 mm. These improvements arise due to shoots partially hidden or poorly illuminated in one view are often better preserved in alternative perspectives. Consequently, measurements across multiple views helps to mitigate random fluctuations caused by local segmentation noise (Wang et al., 2025). Nevertheless, D_P retained a systematic overestimation with a MAE of 4.28 mm, suggesting that multi-view fusion cannot entirely rectify the issue of mask expansion. In contrast, D_R maintained a minimal MAE of 0.83 mm, demonstrating that boundary refinement is essential to suppress systematic bias. Sample-level consistency is visualized in the

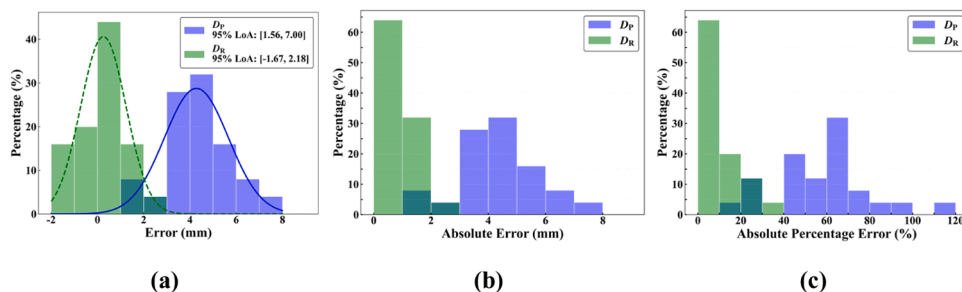


Fig. 11. Error distribution of FS diameters estimation under multi-view observation with (a) error, (b) absolute error, and (c) absolute percentage error.

lollipop chart of Fig. 10, where D_R values closely track the ground-truth diameters, whereas unrefined D_R consistently shows large positive deviations. The observed improvement under both D_P and D_R suggests that the influence of viewpoint-dependent distortion is limited within the adopted close-range imaging configuration.

The statistical reliability of multi-view diameter estimation was further assessed through error distribution and significance analysis. Shapiro-Wilk tests were employed and indicated that the signed errors of D_P and D_R did not show evidence of departure from normality followed a normal distribution with p -values of 0.91 and 0.73, respectively. Comprehensive error statistics are visualized in Fig. 11, where errors of D_P and D_R exhibited bell-shaped profiles consistent with the normality assumption (Fig. 11(a)). In contrast, D_R exhibited a more compact distribution centered near zero, with lower absolute errors than those observed in D_P , as shown in Fig. 11(b) and Fig. 11(c). The systematic bias of D_P and D_R was evaluated using one-sample t -tests and Wilcoxon signed-rank tests against zero (Table 7). One-sample t -tests were conducted to determine whether the mean signed errors of D_P and D_R differed significantly from zero ($t(24) = 15.12$, $p = 9.09 \times 10^{-14}$), indicating a positive systematic bias, whereas no significant deviation was observed for D_R ($t(24) = 1.28$, $p = 0.21$). The Wilcoxon signed-rank tests yielded consistent results, with $p = 5.96 \times 10^{-8}$ for D_P and $p = 0.16$ for D_R . To directly evaluate the effect of boundary refinement, the absolute errors of D_P and D_R were further compared on a sample-by-sample basis. The paired absolute-error differences showed no evidence of departure from normality according to the Shapiro-Wilk test ($p = 0.16$). A paired t -test showed that the absolute error of D_R was lower than that of D_P ($t(24) = 11.52$, $p = 2.90 \times 10^{-11}$), with a mean absolute-error reduction of 3.45 mm and a 95% confidence interval (CI) of 2.83 mm to 4.07 mm. Collectively, these analyses show that boundary refinement reduced the magnitude of the diameter estimation error and removed the systematic positive bias observed in D_P under multi-view evaluation. The residual errors arise from a combination of factors, including stereo depth reconstruction uncertainty, local geometric approximation errors in curved slender structures, and boundary ambiguity under complex canopy conditions. These residual uncertainties were reduced by the coarse-to-fine refinement procedure, although they remain potential sources of error under complex canopy conditions.

4. Conclusions

This study investigated diameter estimation of fine-scale FS by explicitly integrating biological structural priors, geometric modeling, and boundary refinement to provide geometric information for kiwifruit robotic selective pollination. Quantitative analysis of the experimental results demonstrates that the proposed coarse-to-fine strategy outperforms direct mask-based approaches. Specifically, D_P exhibited a pronounced systematic positive bias due to over-segmentation artifacts, which achieved a MAE of 4.44 mm with a MAPE of 60%. In contrast, the application of volatility-guided boundary refinement strategy effectively eliminated these artifacts, whereas D_R obtained a superior MAE of 1.09 mm and reduced the MAPE to 14.68%. Tolerance-based evaluation further reveals the robustness of proposed strategy, achieving an accuracy of 98.9% within a 3 mm threshold to D_R , compared to only 20.2% for D_P at the same level. Multi-view evaluation further indicates that averaging measurements from different perspectives primarily reduces random variability caused by occlusion and local noise, while systematic overestimation persists without boundary refinement. As the proposed framework is specifically designed for robotic selective pollination, the present evaluation is confined to flowering-stage orchard conditions. Future work will further investigate the quantitative relationship between flowering-stage FS diameter and sustainable shoot load under different cultivars and growing conditions to support the conversion of geometric perception into pollination decisions. Overall, the proposed framework provides a method for fine-scale shoot diameter estimation and a perceptual foundation for subsequent load-related decision-

making in robotic selective pollination of kiwifruit.

CRedit authorship contribution statement

Leilei He: Writing – original draft, Methodology, Investigation, Data curation. **Shiao Niu:** Methodology, Data curation, Conceptualization. **Qilin Gong:** Methodology, Conceptualization. **Xiaoyu Ma:** Methodology, Data curation. **Junke Rong:** Methodology, Data curation. **Bryan Gilbert Murengami:** Methodology, Data curation, Conceptualization. **Rui Li:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Shen Hin Lim:** Writing – review & editing, Methodology, Conceptualization. **Longsheng Fu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Gemini for grammar checking to improve the readability of article. After using these servers, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Research Link Provided
[Kiwifruit branch estimation images](#) (Github)

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